

Causal Parameters for Public Health Policy

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Disclaimer: All errors are my own



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¹Footnotes are for asides or references

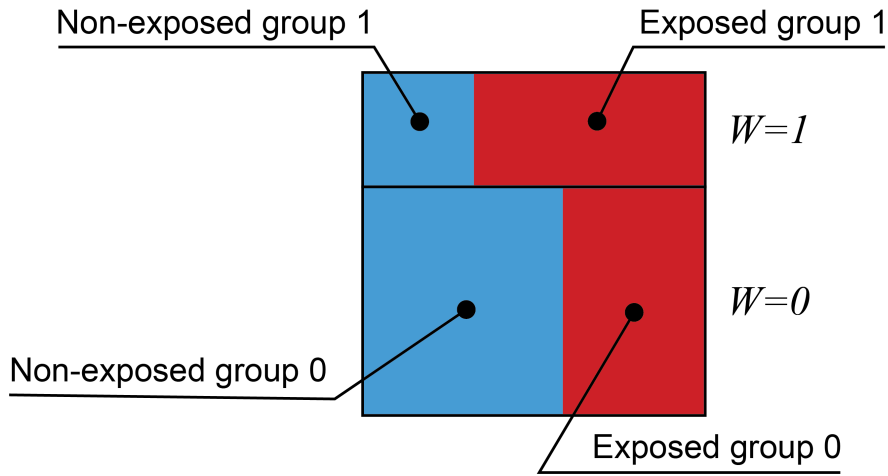
Notation

- Y : outcome
- A : exposure
- W : additional baseline covariates

Perspective: how the tools of causal inference can be used to help generate evidence for the impact of policies

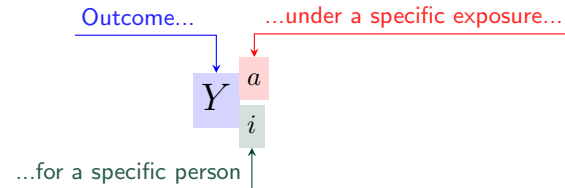
- *Policy*: rule or plan to modify A for a defined group of people
- *Policy Impact*: how the changes in A modify Y (intended or unintended)

Population Diagram Key



Causal Effects

Will define causal effects using potential outcomes



For a binary exposure A , we can think of each person have two potential outcomes

- Potential outcome under exposure Y^1
- Potential outcome under non-exposure Y^0

So causal effects are defined as contrasts of potential outcome

Average Causal Effect – Definition

One of the more well-known estimands from causal inference

$$E[Y_i^{01}] - E[Y_i^{11}]$$

where $E[\cdot]$ is the expected value function²

Contrasts when every person *is not exposed* versus when every person *is exposed*

- This is implicitly what trials estimate

²which you can think about as $1/n \sum_i^n (\cdot)$. Also $E[X] = \Pr(X = 1)$ when X is binary

Average Causal Effect – Diagram



versus



Average Causal Effect – Limitations

The average causal effect may not always be helpful for policy making

Consider tobacco use:

- Scenario where everyone uses tobacco is unlikely to be useful

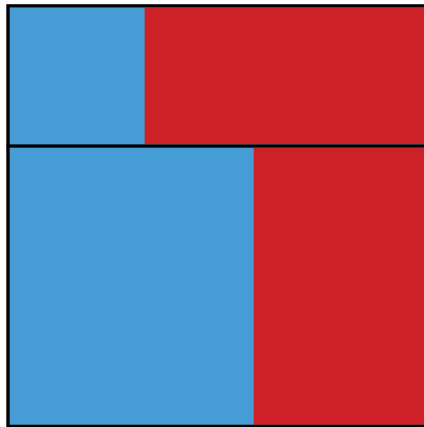
Potential outcomes can be used to define other policies

To help define additional estimands, we introduce the *natural course*³

$$E[Y_i^{A_i}] = E[Y_i]$$

or the outcomes under the exposure distribution as observed

- The population as it is



³Rudolph et al. *Am J Epidemiol* 2022; 191(2):341-348

Removing an Exposure – Definition

Rather than force everyone to be exposed, can compare removal of an exposure⁴

$$E[Y^0] - E[Y]$$

This comparison may have greater utility for harmful exposures

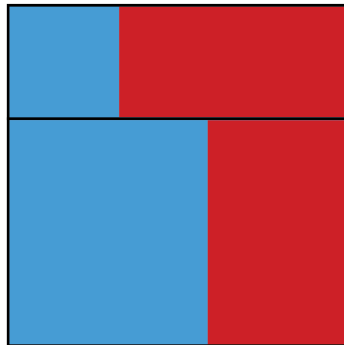
- Ex: reduction in outcome if eliminated all smoking

⁴In the literature, these have been referred to as 'population attributable' effects
Westreich *Epidemiol* 2017; 28(4):525-528

Removing an Exposure – Diagram



versus



Reducing Exposure – Definition

Removing an exposure completely is often overly optimistic

- Preventing *all* tobacco smoking is not currently feasible

Rather than *deterministic* changes in exposure, we can consider changes in the *probability* of exposure

$$\left(\sum_{a \in \{0,1\}} E[Y^a] \times \Pr^*(A = a) \right) - E[Y]$$

where $\Pr^*$ denotes this is specified by the researcher

Reducing Exposure – Definition

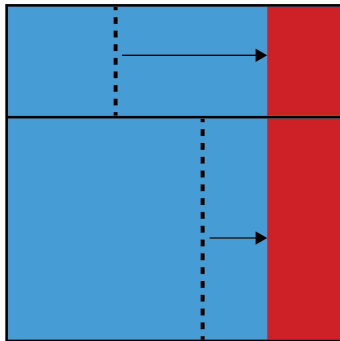
For our binary A , this can also be written as

$$(E[Y^1] \times \Pr^*(A = 1) + E[Y^0] \times \Pr^*(A = 0)) - E[Y]$$

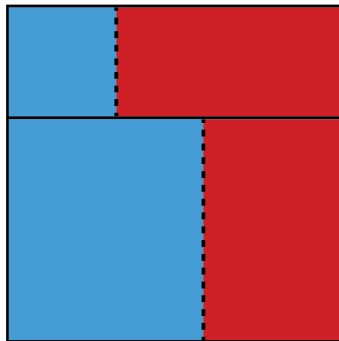
so think about this estimand as a weighted average of the mean potential outcomes, where the weights are the specified policy⁵

⁵Note that the removal of an exposure is the special case where $\Pr^*(A = 0) = 1$

Reducing Exposure – Diagram



versus



Conditional Reductions – Definition

Rather than set everyone to have the same probability of exposure, we can also modify the probability of exposure by W

$$E \left[\sum_{a \in \{0,1\}} Y^a \times \Pr^*(A = a \mid W) \right] - E[Y]$$

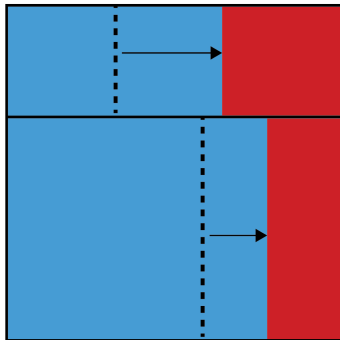
where the policy now incorporates W .⁶

Study policies that differ in effectiveness of changing exposures by groups

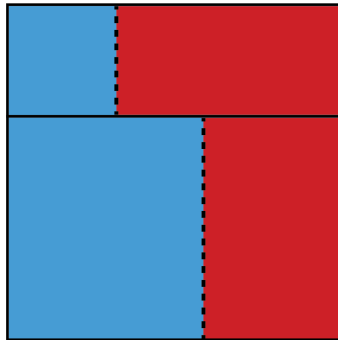
- A smoking cessation program may be more effective in women relative to men

⁶Incremental propensity score effects are one example
Naimi et al. *Epidemiol* 2021;32(2):202-208

Conditionally Reductions – Diagram



versus



Comparing Exposure Reductions – Definition

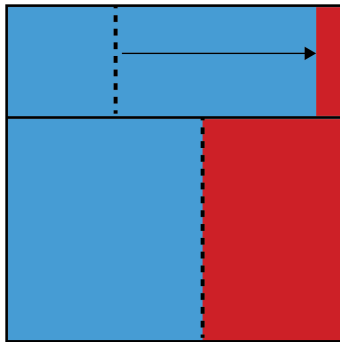
Rather than compare to the natural course, can directly compare competing policies

$$E \left[\sum_{a \in \{0,1\}} Y^a \times \Pr^*(A = a \mid W) \right] - E \left[\sum_{a \in \{0,1\}} Y^a \times \Pr^\#(A = a \mid W) \right]$$

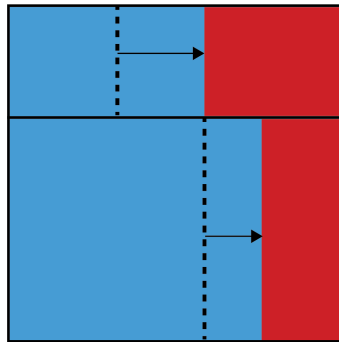
For example, is it more effective to (1) greatly reduce exposure in one high-risk group, or (2) moderately reduce exposure across the population⁷

⁷Drawing inspiration from Rose *Int J Epidemiol* 2001; 30(3):427-432

Comparing Exposure Reductions – Diagram



versus



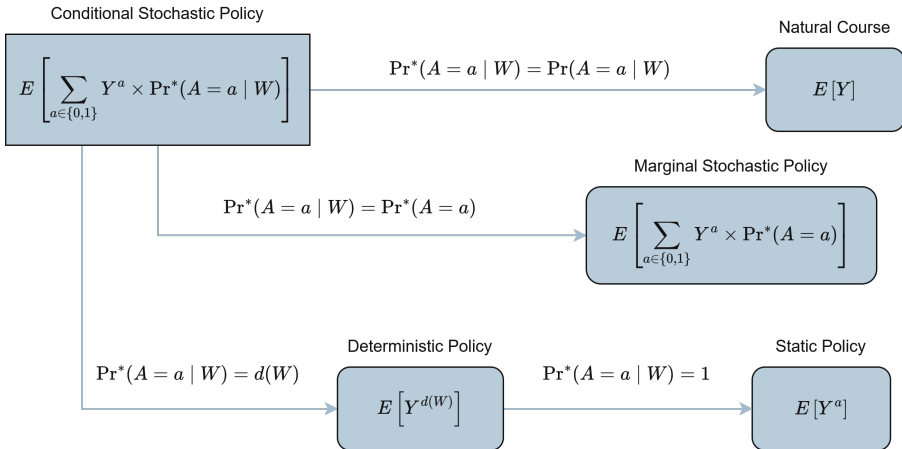
A Unified Expression for Policies

All the previous estimands can be expressed from the conditional reduction formula

$$E \left[\sum_{a \in \{0,1\}} Y^a \times \Pr^*(A = a \mid W) \right]$$

where each of the previous is a different specification for $\Pr^*(A = a \mid W)$

Estimand Flow Diagram



Previous result indicates that identification proceeds similarly for all estimands⁸

Estimation can also be done using g-computation,⁹ or inverse probability weighting

- G-computation: set A according to specified policy and apply as normal
- Inverse probability weighting:

$$I(A = a) \frac{\Pr^*(A = a \mid W)}{\Pr(A = a \mid W)}$$

⁸In fact, identification for probabilistic policies can proceed under weaker assumptions relative to the average causal effect, since exchangeability and positivity may only need to hold for subsets of the covariate space

⁹Ahern et al. *Am J Epidemiol* 2009; 169(9):1140-1147

Potential outcomes aid in the mapping from a specific question to the mathematical machinery that allows for quantitative effect estimation

- Can help to ensure results provide utility for decision makers

Offer the greatest utility when understand how policy acts on A , or $\Pr^*(A = a \mid W)$

- If estimating $\Pr^*(A = a \mid W)$ using another data source, need to be careful about transportability and appropriately estimating the variance¹⁰

¹⁰A data fusion perspective will probably be helpful here
Cole et al. *Am J Epidemiol* 2023; 192(3):467-474

Thank You!



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